Java Interview Preparation

Part 5: Top 100 Pattern Printing Questions

Section 1: Star Patterns (Q1–Q25)

Q1: Right-Angled Triangle Star Pattern (N=5).

```
public static void pattern1(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q2: Inverted Right-Angled Triangle Star Pattern.

```
public static void pattern2(int n) {
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q3: Left-Mirrored Right Triangle Star Pattern.

```
public static void pattern3(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q4: Inverted Left-Mirrored Right Triangle.

```
public static void pattern4(int n) {
    for (int i = n; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q5: Pyramid Star Pattern.

```
public static void pattern5(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= (2 * i - 1); j++) System.out.print("*");
        System.out.println();
    }
}
```

Q6: Inverted Pyramid Star Pattern.

```
public static void pattern6(int n) {
    for (int i = n; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= (2 * i - 1); j++) System.out.print("*");
        System.out.println();
    }
}
```

Q7: Diamond Star Pattern.

```
public static void pattern7(int n) {
    // Upper Pyramid
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
    // Lower Inverted Pyramid
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q8: Half Diamond Star Pattern.

```
public static void pattern8(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int stars = (i <= n) ? i : (2 * n - i);
        for (int j = 1; j <= stars; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q9: Inverted Half Diamond Star Pattern.

```
public static void pattern9(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int stars = (i <= n) ? i : (2 * n - i);
        int spaces = n - stars;
        for (int s = 1; s <= spaces; s++) System.out.print(" ");
        for (int j = 1; j <= stars; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q10: Hollow Square Star Pattern.

```
public static void pattern10(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q11: Hollow Right-Angled Triangle.

```
public static void pattern11(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            if (j == 1 || j == i || i == n) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q12: Hollow Pyramid Star Pattern.

```
public static void pattern12(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1 || i == n) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q13: Hollow Diamond Star Pattern.

```
public static void pattern13(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q14: Hourglass Star Pattern.

```
public static void pattern14(int n) {
    for (int i = n; i >= 1; i--) {
        for (int s = 0; s < n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
    for (int i = 2; i <= n; i++) {
        for (int s = 0; s < n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q15: Butterfly Star Pattern.

```
public static void pattern15(int n) {
    // Upper Part
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print("*");
        for (int s = 1; s <= 2 * (n - i); s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
    // Lower Part
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) System.out.print("*");
        for (int s = 1; s <= 2 * (n - i); s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q16: Rhombus Star Pattern.

```
public static void pattern16(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= n; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q17: Hollow Rhombus Pattern.

```
public static void pattern17(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q18: Diagonal Star Cross (X Pattern).

```
public static void pattern18(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == j || i + j == n + 1) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q19: Plus (+) Star Pattern.

```
public static void pattern19(int n) { // n should be odd
    int mid = n / 2 + 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == mid || j == mid) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q20: Solid Square Star Pattern.

```
public static void pattern20(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q21: Right Arrow Star Pattern.

```
public static void pattern21(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s < i; s++) System.out.print("  ");
        for (int j = i; j <= n; j++) System.out.print("*");
        System.out.println();
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s < i; s++) System.out.print("  ");
        for (int j = i; j <= n; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q22: Left Arrow Star Pattern.

```
public static void pattern22(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= n - i + 1; j++) System.out.print("*");
        System.out.println();
    }
    for (int i = 2; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q23: Swastika Star Pattern.

```
public static void pattern23(int n) { // n must be odd
    int mid = n / 2 + 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == mid || j == mid ||
               (i == 1 && j >= mid) || (i == n && j <= mid) ||
               (j == 1 && i <= mid) || (j == n && i >= mid)) {
                System.out.print("*");
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Q24: Eight / Infinity Star Pattern.

```
public static void pattern24(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        for (int j = 1; j <= n; j++) {
            if ((i == 1 || i == n || i == 2 * n - 1) && (j > 1 && j < n)) System.out.print("*");
            else if ((i > 1 && i < n) || (i > n && i < 2 * n - 1)) {
                if (j == 1 || j == n) System.out.print("*");
                else System.out.print(" ");
            } else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q25: Heart Shape Star Pattern.

```
public static void pattern25(int n) {
    for (int i = n / 2; i <= n; i += 2) {
        for (int j = 1; j < n - i; j += 2) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        for (int j = 1; j <= n - i; j++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
    for (int i = n; i >= 1; i--) {
        for (int j = i; j < n; j++) System.out.print(" ");
        for (int j = 1; j <= (i * 2) - 1; j++) System.out.print("*");
        System.out.println();
    }
}
```

Section 2: Number Patterns (Q26–Q50)

Q26: Simple Number Triangle Pattern.

```
public static void pattern26(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print(j);
        System.out.println();
    }
}
```

Q27: Repeated Number Row Pattern.

```
public static void pattern27(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print(i);
        System.out.println();
    }
}
```

Q28: Inverted Number Triangle Pattern.

```
public static void pattern28(int n) {
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) System.out.print(j);
        System.out.println();
    }
}
```

Q29: Floyd's Triangle Pattern.

```
public static void pattern29(int n) {
    int num = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            System.out.print(num++ + " ");
        }
        System.out.println();
    }
}
```

Q30: 0-1 Binary Triangle Pattern.

```
public static void pattern30(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            if ((i + j) % 2 == 0) System.out.print("1");
            else System.out.print("0");
        }
        System.out.println();
    }
}
```

Q31: Number Pyramid Pattern.

```
public static void pattern31(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print(j + " ");
        System.out.println();
    }
}
```

Q32: Palindromic Number Pyramid Pattern.

```
public static void pattern32(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = i; j >= 1; j--) System.out.print(j);
        for (int j = 2; j <= i; j++) System.out.print(j);
        System.out.println();
    }
}
```

Q33: Pascal's Triangle Pattern.

```
public static void pattern33(int n) {
    for (int i = 0; i < n; i++) {
        for (int s = 0; s < n - i - 1; s++) System.out.print(" ");
        int num = 1;
        for (int j = 0; j <= i; j++) {
            System.out.print(num + " ");
            num = num * (i - j) / (j + 1);
        }
        System.out.println();
    }
}
```

Q34: Continuous Number Pyramid Pattern.

```
public static void pattern34(int n) {
    int val = 1;
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print(val++ + " ");
        System.out.println();
    }
}
```

Q35: Reverse Number Triangle Pattern.

```
public static void pattern35(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = n; j >= i; j--) System.out.print(j);
        System.out.println();
    }
}
```

Q36: Alternate Number Row Pattern.

```
public static void pattern36(int n) {
    int k = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print(k++);
        System.out.println();
    }
}
```

Q37: Diamond Number Pattern.

```
public static void pattern37(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print(j);
        for (int j = i - 1; j >= 1; j--) System.out.print(j);
        System.out.println();
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print(j);
        for (int j = i - 1; j >= 1; j--) System.out.print(j);
        System.out.println();
    }
}
```

Q38: Hollow Number Square Pattern.

```
public static void pattern38(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) System.out.print(j);
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q39: Zig-Zag Number Pattern.

```
public static void pattern39(int n) { // n columns = 9
    for (int i = 1; i <= 3; i++) {
        for (int j = 1; j <= n; j++) {
            if (((i + j) % 4 == 0) || (i == 2 && j % 4 == 0)) {
                System.out.print(j);
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Q40: Concentric Square Number Pattern (Spiral Box Pattern).

```
public static void pattern40(int n) {
    int size = 2 * n - 1;
    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            int top = i, left = j;
            int right = size - 1 - j, bottom = size - 1 - i;
            int minVal = Math.min(Math.min(top, bottom), Math.min(left, right));
            System.out.print(n - minVal);
        }
        System.out.println();
    }
}
```

Q41: Column Increasing Number Triangle.

```
public static void pattern41(int n) {
    for (int i = 1; i <= n; i++) {
        int val = i;
        for (int j = 1; j <= i; j++) {
            System.out.print(val + " ");
            val += (n - j);
        }
        System.out.println();
    }
}
```

Q42: Inverted Odd Number Pyramid.

```
public static void pattern42(int n) {
    for (int i = n; i >= 1; i--) {
        for (int s = 0; s < n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print(2 * j - 1);
        System.out.println();
    }
}
```

Q43: Snake / Serpent Number Matrix Pattern.

```
public static void pattern43(int n) {
    int val = 1;
    for (int i = 1; i <= n; i++) {
        if (i % 2 != 0) {
            for (int j = 1; j <= n; j++) System.out.print(val++ + " ");
        } else {
            int temp = val + n - 1;
            for (int j = 1; j <= n; j++) System.out.print(temp-- + " ");
            val += n;
        }
        System.out.println();
    }
}
```

Q44: Cross Diagonal Number Pattern.

```
public static void pattern44(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == j) System.out.print(i);
            else if (i + j == n + 1) System.out.print(j);
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q45: Increasing-Decreasing Row Number Triangle.

```
public static void pattern45(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print(j);
        for (int j = i - 1; j >= 1; j--) System.out.print(j);
        System.out.println();
    }
}
```

Q46: Binary Alternate Column Pattern.

```
public static void pattern46(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) System.out.print(j % 2);
        System.out.println();
    }
}
```

Q47: Binary Alternate Row Pattern.

```
public static void pattern47(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) System.out.print(i % 2);
        System.out.println();
    }
}
```

Q48: Number Hourglass Pattern.

```
public static void pattern48(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s < i; s++) System.out.print(" ");
        for (int j = i; j <= n - i + 1; j++) System.out.print(j);
        System.out.println();
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s < i; s++) System.out.print(" ");
        for (int j = i; j <= n - i + 1; j++) System.out.print(j);
        System.out.println();
    }
}
```

Q49: Fibonacci Series Triangle Pattern.

```
public static void pattern49(int n) {
    int a = 0, b = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
        System.out.println();
    }
}
```

Q50: Multiplication Table Grid Pattern (N x N).

```
public static void pattern50(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            System.out.printf("%2d ", i * j);
        }
        System.out.println();
    }
}
```

Section 3: Alphabet / Character Patterns (Q51–Q75)

Q51: Right-Angled Character Triangle Pattern.

```
public static void pattern51(int n) {
    for (int i = 1; i <= n; i++) {
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q52: Repeated Character Row Pattern.

```
public static void pattern52(int n) {
    for (int i = 0; i < n; i++) {
        char ch = (char) ('A' + i);
        for (int j = 0; j <= i; j++) System.out.print(ch);
        System.out.println();
    }
}
```

Q53: Inverted Character Triangle.

```
public static void pattern53(int n) {
    for (int i = n; i >= 1; i--) {
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q54: Continuous Alphabet Triangle (Floyd's Character Pattern).

```
public static void pattern54(int n) {
    char ch = 'A';
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) System.out.print(ch++ + " ");
        System.out.println();
    }
}
```

Q55: Character Pyramid Pattern.

```
public static void pattern55(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        for (char ch = (char) ('A' + i - 2); ch >= 'A'; ch--) System.out.print(ch);
        System.out.println();
    }
}
```

Q56: Reverse Character Triangle Pattern.

```
public static void pattern56(int n) {
    for (int i = 0; i < n; i++) {
        for (char ch = (char) ('A' + n - 1); ch >= 'A' + n - 1 - i; ch--) {
            System.out.print(ch);
        }
        System.out.println();
    }
}
```

Q57: K-Shape Alphabet Pattern.

```
public static void pattern57(int n) {
    for (int i = n; i >= 1; i--) {
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        System.out.println();
    }
    for (int i = 2; i <= n; i++) {
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q58: Diamond Character Pattern.

```
public static void pattern58(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        for (char ch = (char) ('A' + i - 2); ch >= 'A'; ch--) System.out.print(ch);
        System.out.println();
    }
    for (int i = n - 1; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        for (char ch = (char) ('A' + i - 2); ch >= 'A'; ch--) System.out.print(ch);
        System.out.println();
    }
}
```

Q59: Hollow Character Square.

```
public static void pattern59(int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i == 0 || i == n - 1 || j == 0 || j == n - 1) {
                System.out.print((char) ('A' + j));
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Q60: Character Cross (X) Pattern.

```
public static void pattern60(int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i == j || i + j == n - 1) System.out.print((char) ('A' + j));
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q61: Lowercase Right Triangle Pattern.

```
public static void pattern61(int n) {
    for (int i = 1; i <= n; i++) {
        for (char ch = 'a'; ch < 'a' + i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q62: Descending Alphabet Triangle.

```
public static void pattern62(int n) {
    for (int i = 0; i < n; i++) {
        for (char ch = (char) ('E' - i); ch <= 'E'; ch++) System.out.print(ch + " ");
        System.out.println();
    }
}
```

Q63: Alphabet Hourglass Pattern.

```
public static void pattern63(int n) {
    for (int i = 0; i < n; i++) {
        for (int s = 0; s < i; s++) System.out.print(" ");
        for (char ch = (char) ('A' + i); ch <= 'A' + n - 1 - i; ch++) System.out.print(ch);
        System.out.println();
    }
    for (int i = n - 2; i >= 0; i--) {
        for (int s = 0; s < i; s++) System.out.print(" ");
        for (char ch = (char) ('A' + i); ch <= 'A' + n - 1 - i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q64: Alphabet Right-Angled Mirror Pattern.

```
public static void pattern64(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q65: Alternate Case Alphabet Pattern.

```
public static void pattern65(int n) {
    for (int i = 1; i <= n; i++) {
        boolean upper = (i % 2 != 0);
        for (int j = 0; j < i; j++) {
            char ch = upper ? (char) ('A' + j) : (char) ('a' + j);
            System.out.print(ch + " ");
        }
        System.out.println();
    }
}
```

Q66: Alphabet Rhombus Pattern.

```
public static void pattern66(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + n; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q67: Word Printing Repeated Pattern (e.g., "JAVA").

```
public static void pattern67(String word) {
    for (int i = 1; i <= word.length(); i++) {
        System.out.println(word.substring(0, i));
    }
}
```

Q68: Word Printing Square Matrix.

```
public static void pattern68(String word) {
    for (int i = 0; i < word.length(); i++) {
        System.out.println(word);
    }
}
```

Q69: Character Butterfly Pattern.

```
public static void pattern69(int n) {
    for (int i = 1; i <= n; i++) {
        for (char ch = 'A'; ch < 'A' + i; ch++) System.out.print(ch);
        for (int s = 1; s <= 2 * (n - i); s++) System.out.print(" ");
        for (char ch = (char) ('A' + i - 1); ch >= 'A'; ch--) System.out.print(ch);
        System.out.println();
    }
}
```

Q70: Alphabet Snake Matrix Pattern.

```
public static void pattern70(int n) {
    char ch = 'A';
    for (int i = 1; i <= n; i++) {
        if (i % 2 != 0) {
            for (int j = 1; j <= n; j++) System.out.print(ch++ + " ");
        } else {
            char temp = (char) (ch + n - 1);
            for (int j = 1; j <= n; j++) System.out.print(temp-- + " ");
            ch += n;
        }
        System.out.println();
    }
}
```

Q71: Right Pascal Character Triangle.

```
public static void pattern71(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int cols = (i <= n) ? i : (2 * n - i);
        for (char ch = 'A'; ch < 'A' + cols; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q72: Left Pascal Character Triangle.

```
public static void pattern72(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int cols = (i <= n) ? i : (2 * n - i);
        for (int s = 1; s <= n - cols; s++) System.out.print(" ");
        for (char ch = 'A'; ch < 'A' + cols; ch++) System.out.print(ch);
        System.out.println();
    }
}
```

Q73: Hollow Character Pyramid Pattern.

```
public static void pattern73(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1 || i == n) System.out.print((char) ('A' + i - 1));
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q74: Character Plus (+) Sign Pattern.

```
public static void pattern74(int n) { // n must be odd
    int mid = n / 2 + 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == mid || j == mid) System.out.print("C");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q75: Alphabet Concentric Box Pattern.

```
public static void pattern75(int n) {
    int size = 2 * n - 1;
    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            int top = i, left = j;
            int right = size - 1 - j, bottom = size - 1 - i;
            int minVal = Math.min(Math.min(top, bottom), Math.min(left, right));
            System.out.print((char) ('A' + n - 1 - minVal));
        }
        System.out.println();
    }
}
```

Section 4: Advanced, Special & Symbolic Patterns (Q76–Q100)

Q76: Star-Number Mixed Right Triangle.

```
public static void pattern76(int n) {
    for (int i = 1; i <= n; i++) {
        System.out.print(i);
        for (int j = 1; j <= i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q77: Number-Star Pyramid.

```
public static void pattern77(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j < i; j++) System.out.print("*");
        System.out.print(i);
        for (int j = 1; j < i; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q78: Inverted Star Pyramid with Border Numbers.

```
public static void pattern78(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s < i; s++) System.out.print(" ");
        System.out.print(i);
        if (i != n) {
            for (int j = 1; j <= 2 * (n - i) - 1; j++) System.out.print("*");
            System.out.print(i);
        }
        System.out.println();
    }
}
```

Q79: Hollow Diamond with Inside Number.

```
public static void pattern79(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        System.out.print("*");
        if (i > 1) {
            for (int j = 1; j <= 2 * i - 3; j++) {
                if (j == i - 1) System.out.print(i - 1);
                else System.out.print(" ");
            }
            System.out.print("*");
        }
        System.out.println();
    }
}
```

Q80: Checkerboard / Chessboard Grid Pattern.

```
public static void pattern80(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if ((i + j) % 2 == 0) System.out.print("#");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q81: Concentric Square Hash & Star Border Pattern.

```
public static void pattern81(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) System.out.print("*");
            else System.out.print("#");
        }
        System.out.println();
    }
}
```

Q82: Double Hill Pyramid Pattern.

```
public static void pattern82(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        for (int s = 1; s <= 2 * (n - i) + 1; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q83: Triple Hill Pyramid Pattern.

```
public static void pattern83(int n) {
    for (int i = 1; i <= n; i++) {
        for (int k = 0; k < 3; k++) { // 3 Hills
            for (int s = 1; s <= n - i; s++) System.out.print(" ");
            for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
            for (int s = 1; s <= n - i; s++) System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q84: Trapezoid Number Pattern.

```
public static void pattern84(int n) {
    int lTerm = 1, rTerm = n * n + 1;
    for (int i = n; i >= 1; i--) {
        for (int s = 0; s < n - i; s++) System.out.print("  ");
        for (int j = 1; j <= i; j++) {
            System.out.print(lTerm++ + "*");
        }
        for (int j = 1; j <= i; j++) {
            System.out.print(rTerm++);
            if (j < i) System.out.print("*");
        }
        rTerm -= (2 * i - 2);
        System.out.println();
    }
}
```

Q85: Sandglass Number-Star Mixed Pattern.

```
public static void pattern85(int n) {
    for (int i = n; i >= 1; i--) {
        for (int s = 0; s < n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) {
            System.out.print(j);
            if (j < i) System.out.print("*");
        }
        System.out.println();
    }
    for (int i = 2; i <= n; i++) {
        for (int s = 0; s < n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) {
            System.out.print(j);
            if (j < i) System.out.print("*");
        }
        System.out.println();
    }
}
```

Q86: Hollow Inverted Right Triangle Star Pattern.

```
public static void pattern86(int n) {
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            if (i == n || j == 1 || j == i) System.out.print("*");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q87: Star Box with Cross (Window Frame Pattern).

```
public static void pattern87(int n) { // n odd
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n || i == j || i + j == n + 1) {
                System.out.print("*");
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Q88: Stairs Pattern.

```
public static void pattern88(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) System.out.print("#");
        System.out.println();
    }
}
```

Q89: Wave Pattern.

```
public static void pattern89(int waveLength, int waveHeight) {
    for (int i = 1; i <= waveHeight; i++) {
        for (int j = 1; j <= waveLength; j++) {
            if ((i + j) % 4 == 0 || (i == 2 && j % 4 == 0)) System.out.print("^");
            else System.out.print(" ");
        }
        System.out.println();
    }
}
```

Q90: Binary Inverted Right Triangle.

```
public static void pattern90(int n) {
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) System.out.print(j % 2);
        System.out.println();
    }
}
```

Q91: Left-Aligned Decreasing Arrowhead Pattern.

```
public static void pattern91(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int stars = (i <= n) ? i : (2 * n - i);
        for (int j = 1; j <= stars; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q92: Right-Aligned Arrowhead Pattern.

```
public static void pattern92(int n) {
    for (int i = 1; i <= 2 * n - 1; i++) {
        int stars = (i <= n) ? i : (2 * n - i);
        for (int s = 1; s <= n - stars; s++) System.out.print(" ");
        for (int j = 1; j <= stars; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q93: Alphabet-Star Alternating Pyramid.

```
public static void pattern93(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= i; j++) {
            if (i % 2 != 0) System.out.print((char) ('A' + j - 1) + " ");
            else System.out.print("* ");
        }
        System.out.println();
    }
}
```

Q94: Spiral Number Box Pattern (N x N).

```
public static void pattern94(int n) {
    int size = 2 * n;
    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            int val = Math.min(Math.min(i, j), Math.min(size - 1 - i, size - 1 - j));
            System.out.print((val + 1) + " ");
        }
        System.out.println();
    }
}
```

Q95: Symmetric Double Diamond Pattern.

```
public static void pattern95(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        for (int s = 1; s <= 2 * (n - i) + 1; s++) System.out.print(" ");
        for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
        System.out.println();
    }
}
```

Q96: Number Square with Decreasing Corner Values.

```
public static void pattern96(int n) {
    int size = 2 * n - 1;
    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            int min = Math.min(Math.min(i, j), Math.min(size - 1 - i, size - 1 - j));
            System.out.print(n - min);
        }
        System.out.println();
    }
}
```

Q97: Vertical Striped Star-Space Pattern.

```
public static void pattern97(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) System.out.print("* ");
        System.out.println();
    }
}
```

Q98: Diagonal Line Pattern.

```
public static void pattern98(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s < i; s++) System.out.print(" ");
        System.out.println("*");
    }
}
```

Q99: Anti-Diagonal Line Pattern.

```
public static void pattern99(int n) {
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) System.out.print(" ");
        System.out.println("*");
    }
}
```

Q100: Giant Cross Star Grid Frame.